

Lab Work 04

Topic	Abstraction, Interfaces (using ABCs in Python)
Objective	Understand and apply abstraction by defining and using Abstract Base Classes (ABCs) to enforce contracts (interfaces).
AI Copilot Usage	Not Allowed

Tasks

1. Review ABCs:

- Discuss with the tutor/peers: What is the purpose of the **@abstractmethod** decorator used in Session 3? Why couldn't we create an instance of the **Person** class directly? How does this relate to abstraction?

2. Define **Enrollable** Interface (ABC):

- Define a new ABC named **Enrollable** using the **abc** module.
- Give it one abstract method: **add_student(self, student)** which should take a **Student** object.
- Give it another abstract method: **get_enrollment_count(self)** which should return the number of enrolled students.
- Modify the **Unit** class (from Session 1/2) to inherit from **Enrollable**.
- Implement the **add_student** method in **Unit**. It should store enrolled students in a list (e.g., **self.enrolled_students**). Add basic checks (e.g., prevent adding non-Student objects, prevent duplicates).
- Implement the **get_enrollment_count** method in **Unit**.

3. Use the Interface:

- Create a function **register_student_for_unit(student, unit)** outside the classes.
- Inside this function, check if the **unit** parameter is an instance of the **Enrollable** ABC (**isinstance(unit, Enrollable)**).
- If it is, call the **unit.add_student(student)** method. If not, print an error.
- Test this function by passing a **Student** object and a **Unit** object.
- Try passing a **Student** object and a **Teacher** object (which is not **Enrollable**) to see the error message.
- Instantiate several **Unit** objects, add students using the function, and print the enrollment count for each unit using **get_enrollment_count**.

4. Creative Challenge (Optional, if time permits):

- a) Define a **Reportable** interface (ABC) with an abstract method **generate_report(self)** that should return a string containing a formatted report.
- b) Make the **Unit** class also inherit from **Reportable**. Implement **generate_report** in **Unit** to list the unit code, name, and the names/IDs of enrolled students.
- c) Create a function **print_system_reports(reportable_items)** that takes a list of objects, checks if each item is **Reportable**, and calls **generate_report** if it is. Test this with your **Unit** objects.

Important Notes

- This is an individual task. All code must be your own original work.
- Ask questions and seek help from the lecturer or teaching assistants if you are stuck.
- Focus on writing clean, well-organized, and well-documented code.